

Olist E-Commerce: Sales & Delivery Efficiency Analysis

Business Problem:

E-commerce companies often face two key challenges:

1. Understanding revenue drivers

- Which states, cities, and product categories generate the most revenue?
- What drives high order volume vs high-value sales?

2. Improving customer experience

- How do delivery delays impact customer satisfaction?
- What factors influence freight cost and delivery performance?

The goal of this project is to:

- Analyze sales performance
- Evaluate delivery efficiency
- Identify factors affecting customer satisfaction
- Provide actionable business recommendations

The Solution Overview:

To solve this, I architected a complete end-to-end data analytics pipeline. Rather than just visualizing clean data, I extracted and transformed raw datasets using Python (Pandas), engineered a consolidated relational database schema in MySQL, and developed a highly interactive, dynamic dashboard in Power BI using advanced DAX modelling.

The Process:

Situation

The company lacks clarity on:

- Revenue distribution

- Customer satisfaction drivers
- Delivery performance issues

Task

- Analyze sales and logistics data
- Identify key revenue drivers
- Evaluate impact of delivery delays
- Provide actionable insights

Action

- Cleaned and transformed raw dataset using Python
- Created meaningful features like review categories
- Built structured database in MySQL
- Designed an interactive Power BI dashboard
- Performed analysis on:
 - Revenue trends
 - Order distribution
 - Delivery delays
 - Customer reviews
 - Freight vs product weight

Result (Key Insights):

Sales Insights

- Total Orders: 113K
- Total Customers: 96K
- Total Revenue: \$20.34M
- Average Order Value: \$180.25
- 2018 generated 54.5% of total revenue

- São Paulo (SP) is the top state:
 - Revenue: \$7.6M
 - Orders: 48K
- Top categories by revenue:
 - bed_bath_table
 - health_beauty
 - computers_accessories
- Demand vs Value Insight:
 - furniture_decor has more orders
 - computers_accessories generates more revenue

Indicates higher pricing or premium products

- Peak sales months:
 - March → August
 - Highest in May (\$2.2M)
- Sales drop:
 - September → December

Logistics & Customer Insights

- Avg Delivery Time: 12.41 days
- Avg Freight Value: \$19.99
- Delayed Orders: 7.74%
- Delivery Performance Impact:
 - On-time → 78% positive reviews
 - Delayed → 53% negative reviews

Strong evidence that delivery performance directly impacts customer satisfaction

Freight & Weight Analysis

- Weak positive relationship between:
 - Product weight
 - Freight value
 - Conclusion:
 - Weight has limited impact
 - Other factors (distance, logistics) are more important
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Regional Logistics Insight

- Highest freight cost:
 - RR (\$43), PB (\$43), RO (\$41)
 - Lowest:
 - SP (\$15)
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Delay Drivers

- Higher weight → higher delay probability
- Indicates operational inefficiency for heavy products

Conclusion

This analysis shows that:

- Revenue is concentrated in a few key states and categories
- Customer satisfaction is highly sensitive to delivery performance
- Freight cost is influenced by multiple factors, not just weight
- Operational inefficiencies (delays) significantly impact customer reviews

Final Business Recommendations:

1. Improve Delivery Performance

- Focus on reducing delays
- Optimize logistics in high-delay regions
- Partner with better delivery providers

This will directly improve customer satisfaction

2. Optimize High-Demand Categories

- Increase inventory for top-selling categories
- Promote high-value categories like computers_accessories

3. Focus on Peak Season

- Increase marketing & inventory from March–August
- Prepare for demand spikes

4. Reduce Delivery Delays for Heavy Products

- Improve handling & shipping strategy for heavy items
- Optimize packaging and logistics routes

5. Customer Experience Strategy

- Prioritize fast delivery
- Use delivery performance as a KPI

Since delays lead to 4x more negative reviews

Final Note:

This project demonstrates:

- End-to-end data analysis pipeline
- Business-focused thinking
- Ability to connect data with real-world decisions